

FEA of Very Large Bone Models Based on Micro-CT Scans

A matrix-free, voxel-based μ FE framework in MFEM — scaling to 800 million degrees of freedom

Shani Martinez-Weissberg¹ · Will Pazner² · Zohar Yosibash¹

¹ Tel-Aviv University, School of Mechanical Engineering ² Portland State University, Dept. of Mathematics and Statistics

Predicting Bone Fracture Risk Following LITT

- Global bone cancer treatment market: \$1.45B global market (2026), growing at ~5.6% CAGR.
- Preclinical treatment: Laser-induced thermotherapy (LITT).
- Preclinical animal model: New Zealand White (NZW) rabbits.
- High-resolution imaging: μ CT at 20–40 μ m resolution.

Our goal

Validated μ FE models for predicting bone fracture risk following LITT

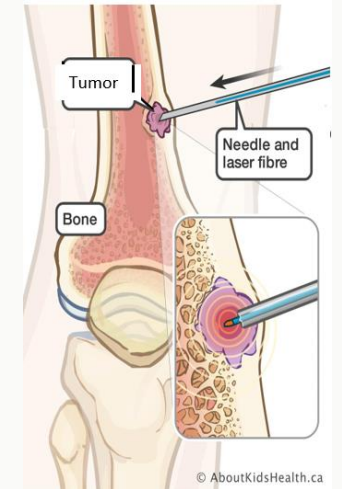
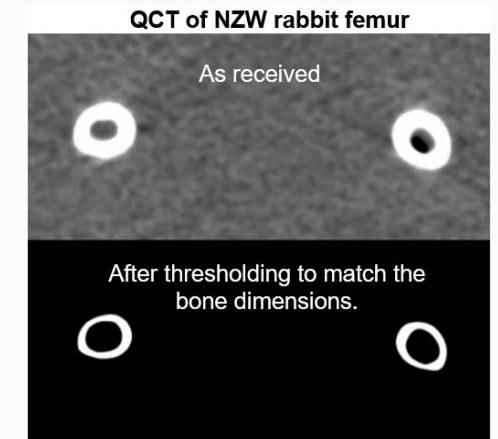
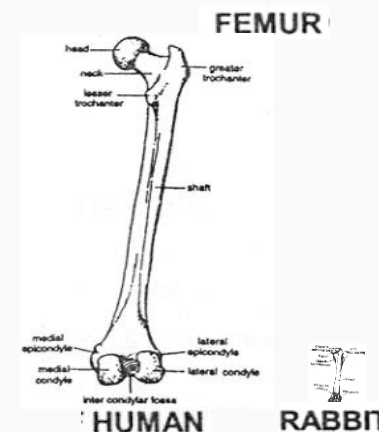
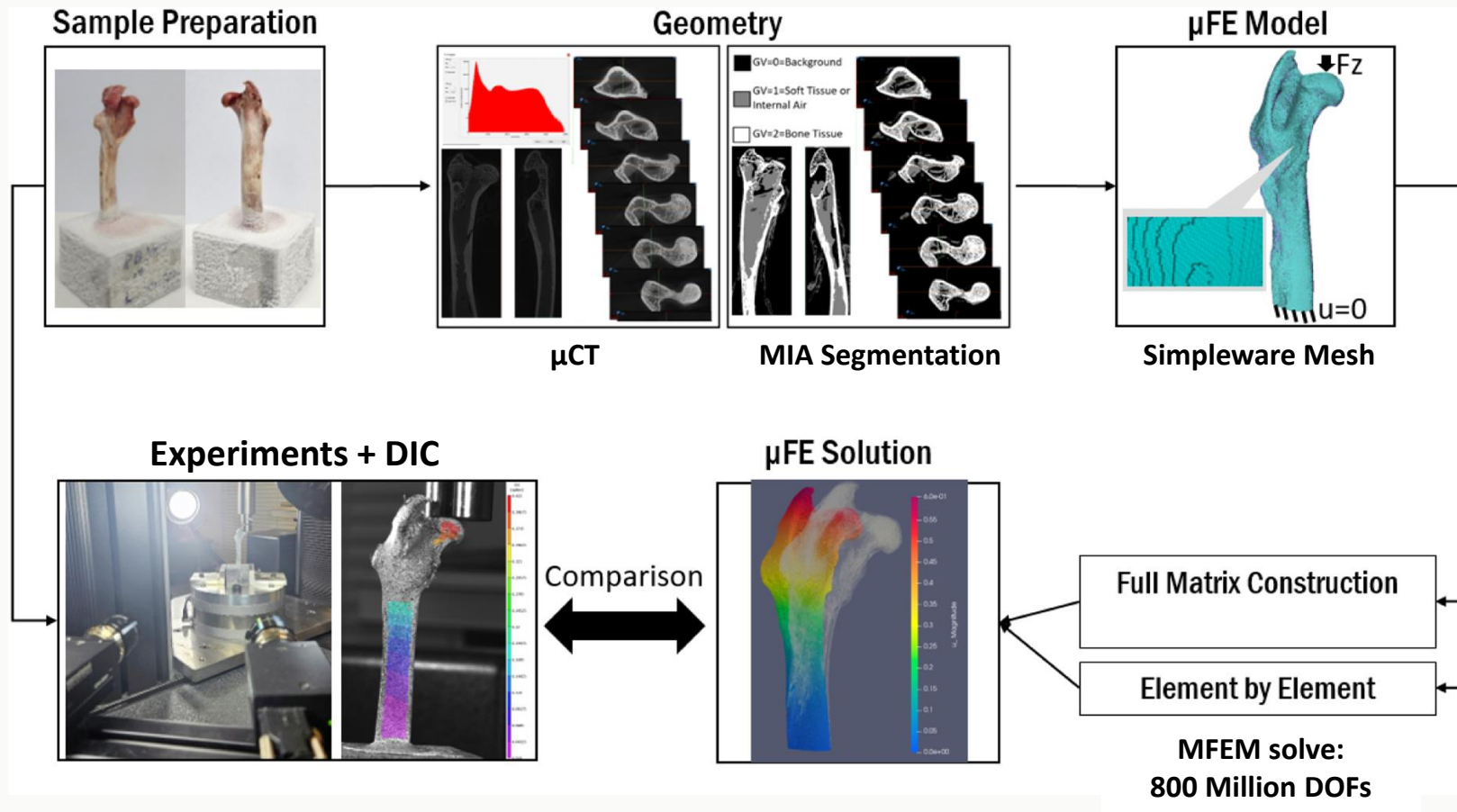


Illustration © AboutKidsHealth.ca, The Hospital for Sick Children (SickKids). Used for illustrative purposes.



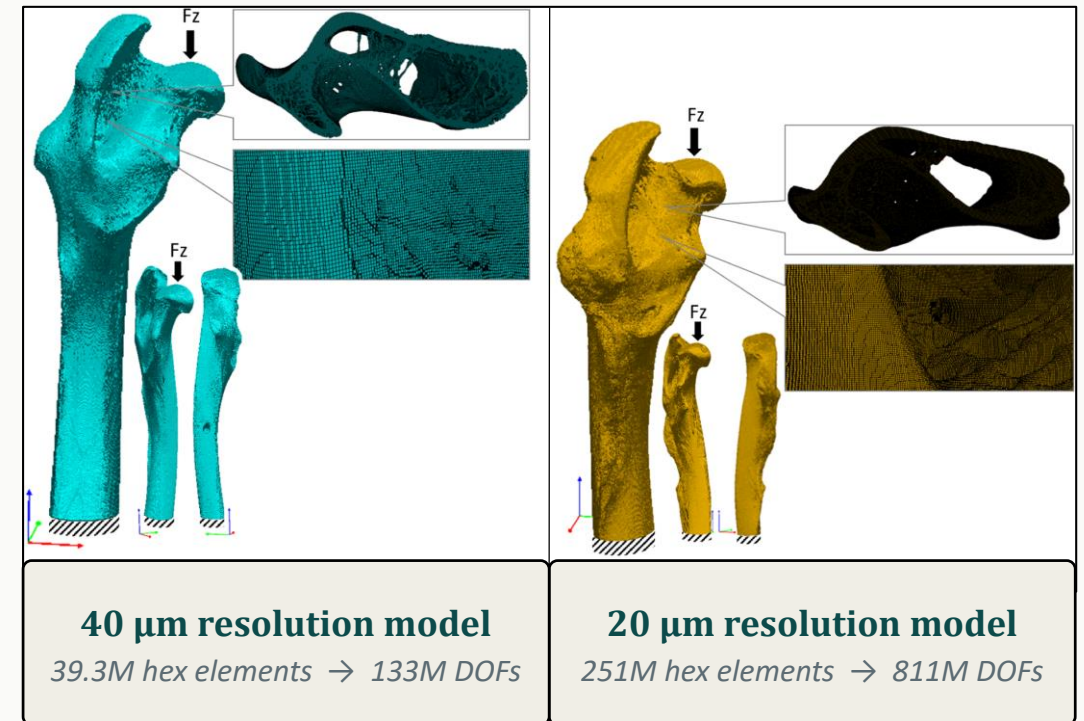
End-to-End Pipeline: μ CT to Validated μ FE Solution



The μ FE Model with MFEM

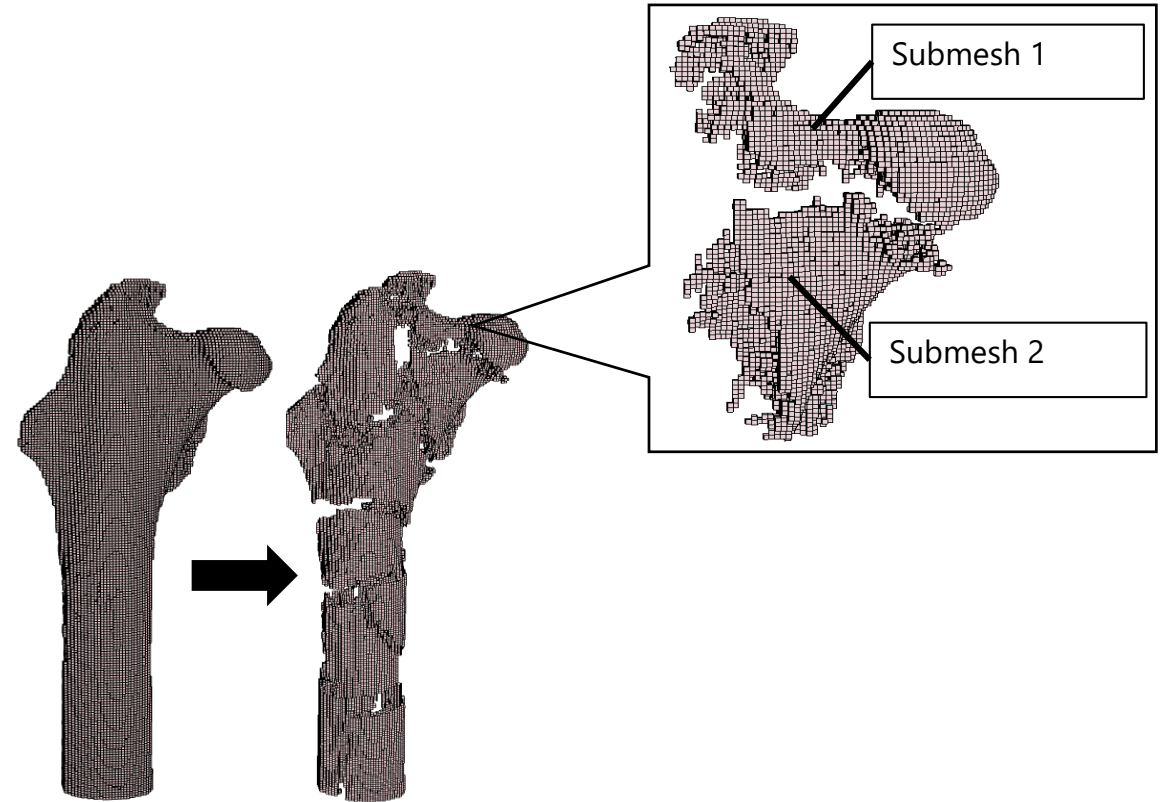
- Software: MFEM.
- FE linear elastic problem: Find $\mathbf{u} \in [H_{\Gamma_D}^1(\Omega)]^3$ such that,

$$\int_{\Omega} [2\mu\boldsymbol{\varepsilon}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v}) + \lambda \operatorname{div}\mathbf{u} \operatorname{div}\mathbf{v}] d\Omega = \int_{\Gamma_N} \mathbf{T}^T \mathbf{v} d\Gamma \text{ for all } \mathbf{v} \in [H_{\Gamma_D}^1(\Omega)]^3$$
- Discretization to a linear system: $A\bar{\mathbf{x}} = \bar{\mathbf{b}}$
- Boundary conditions:
 Bottom surface: $\bar{u} = 0$;
 Surface traction on femur's head: $T_z = -235 \text{ MPa} \rightarrow F_z = -100 \text{ N}$.
- **Assumed** homogeneous isotropic material properties:
 $E=10 \text{ GPa}, \nu=0.3$.
- Linear 8-node hexahedral elements
- Huge FE model



Handling Huge FE models $\sim 10^9$ DOF with MFEM

- Solving iterative method for $A\bar{x} = \bar{f}$: Preconditioned Conjugate Gradient (PCG)
- **Mesh Partition** into subdomains across MPI ranks (METIS).
- **Decompose operator** $A = P^T G^T \hat{A} G P$ where A -global sparse matrix and $\hat{A}$ - block element-local operator

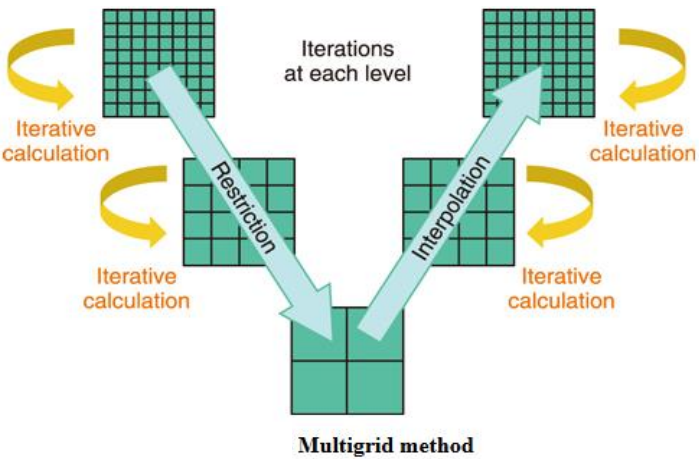


George Karypis and Vipin Kumar “A Fast and Highly Quality Multilevel Scheme for Partitioning Irregular Graphs” SIAM Journal on Scientific Computing, 1999 , 20, No. 1, 359—392.

Handling Huge FE models $\sim 10^9$ DOF with MFEM

- Two ways to apply Ax :
 - I. Conventional method: Full Assembly (FA)
 - II. Partial Assembly: Element by element (EbE)

	FA	EBE
Iterative solver	PCG	PCG
Global stiffness matrix stored?	Yes	No
Convergence	Fewer iterations	More iterations
Memory	Higher	Lower

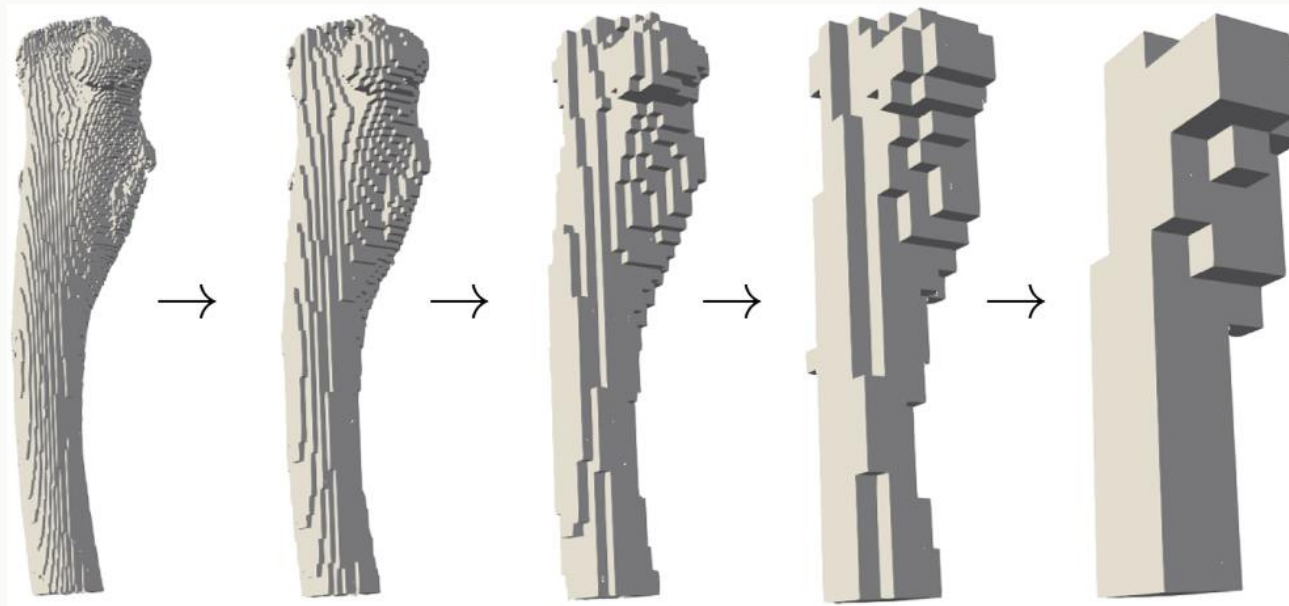


- Multigrid preconditioner: Algebraic Multigrid (AMG) & Geometric Voxel Multigrid (VMG)
- Removing the matrix solves the memory problem — but it also removes the matrix that AMG normally needs.

	FA	EBE
preconditioner	AMG	VMG
Multigrid hierarchy	Coarse matrices	Coarse meshes

Voxel Multigrid (VMG) Preconditioner

μ CT voxel mesh = finest level $\rightarrow$ successive $2 \times 2 \times 2$ voxel coarsening $\rightarrow$ semi-structured multigrid hierarchy
(loss of fine details)



Fine mesh $\rightarrow$ successive semi-structured voxel coarsening (Algorithm 1)

Voxel Multigrid (VMG) Preconditioner Smoothing

- Standard: Multigrid needs a cheap smoother at every level to rapidly damp high-frequency error.

	FA	EBE
Stores A?	Yes	No, only $\hat{A}$
Smoother	Gauss–Seidel/ ILU/ etc.	Block Jacobi + Chebyshev
Diagonal D (what the smoother actually uses)	Read straight off the stored A	Computed matrix-free from element-local blocks $\hat{A}$

- Chebyshev acceleration: each Jacobi step uses a different, precomputed weight — chosen to damp high-frequency error as fast as mathematically possible.

Matrix-free operator

+

Matrix-free multigrid preconditioner

Voxel Multigrid (VMG) vs. Algebraic Multigrid (AMG)

– Test case: reduced femur (72,070 hex elements) with pure-displacement BCs uniformly refined

CG iterations: comparable · **Assembly:** VMG dramatically faster · **Setup:** VMG faster · **Solve:** comparable · **Memory:** VMG needs less at scale

ℓ	# DOFs	Method	CG It.	Assembly (s)	Setup (s)	Solve (s)	Memory (GB)
Refinement ↓	0	AMG	6	1.73×10^{-1}	4.68×10^{-2}	6.15×10^{-2}	23.9
		VMG	6	1.07×10^{-4}	3.39×10^{-2}	6.52×10^{-2}	23.4
	1	AMG	7	1.56×10^0	4.54×10^{-1}	8.29×10^{-1}	29.0
		VMG	6	2.09×10^{-4}	1.82×10^{-1}	6.29×10^{-1}	24.7
	2	AMG	7	1.66×10^1	4.54×10^0	8.24×10^0	66.0
		VMG	6	1.14×10^{-4}	1.55×10^0	6.91×10^0	33.9
	3	AMG	8	2.98×10^2	5.91×10^1	9.62×10^1	354.6
		VMG	6×10^6	3.48×10^{-4}	1.31×10^1	5.58×10^1	105.4

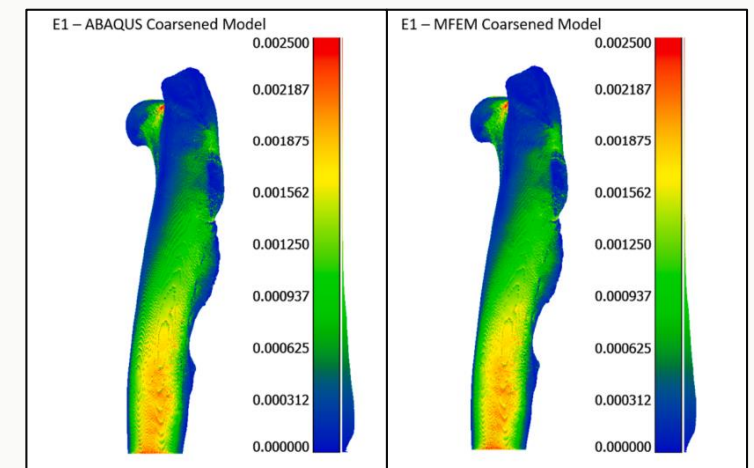
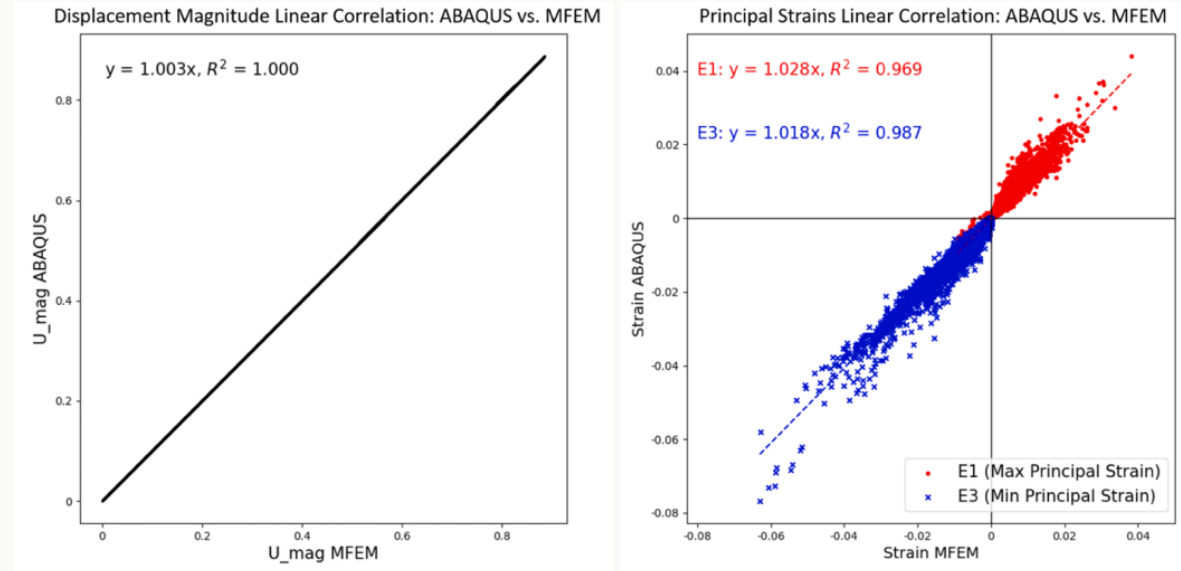
VERIFICATION

MFEM vs. the Commercial Solver Abaqus

- **L20 resampled to 80 μm :** 3.59M hex elements · ~14M DOFs
- Material properties: $E = 10 \text{ GPa}$, $\nu = 0.3$
- Boundary conditions: vertical force of 91.4 N at the femoral head, distal end clamped $u=0$.
- Same model solved by both Abaqus and MFEM-FA on 12 processors
- 4,646,535 nodal displacements and principal strains compared point-by-point

Verification

MFEM's FA formulation is correct on the bone geometry — clearing the way for the full-scale, uncoarsened models.

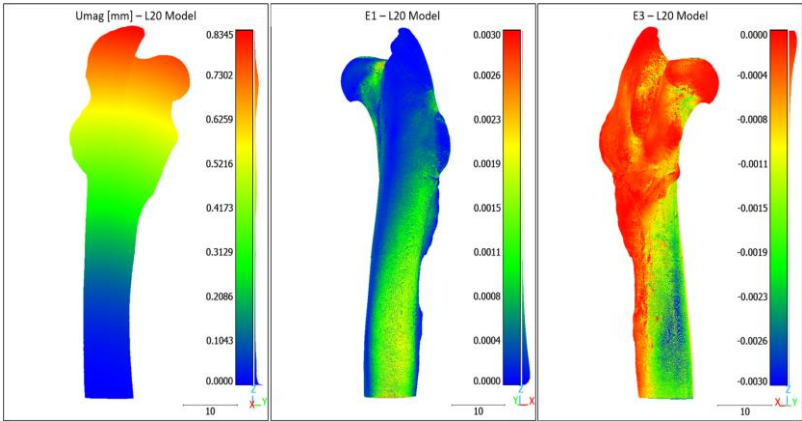


Full Assembly (FA) vs. Element-by-Element (EBE)

– Identical model · 5 nodes · 192 MPI ranks · mesh partitioned with METIS¹

Model	DOFs	Method	CG It.		Solve (min)		Memory (GB)	
L40	133M	FA	× 2.3 ↺	302	× 2.6 ↺	11.4	× 3.1 ↺	499
		EBE		709		17.5		159
L20	811M	FA	× 2.6 ↺	840	× 2.6 ↺	245	× 4.1 ↺	2818
		EBE		2218		399		692

- EBE trades runtime for a much smaller memory footprint
- **Identical solutions:** displacement magnitude and principal strains ($R^2 \approx 1.000$) .

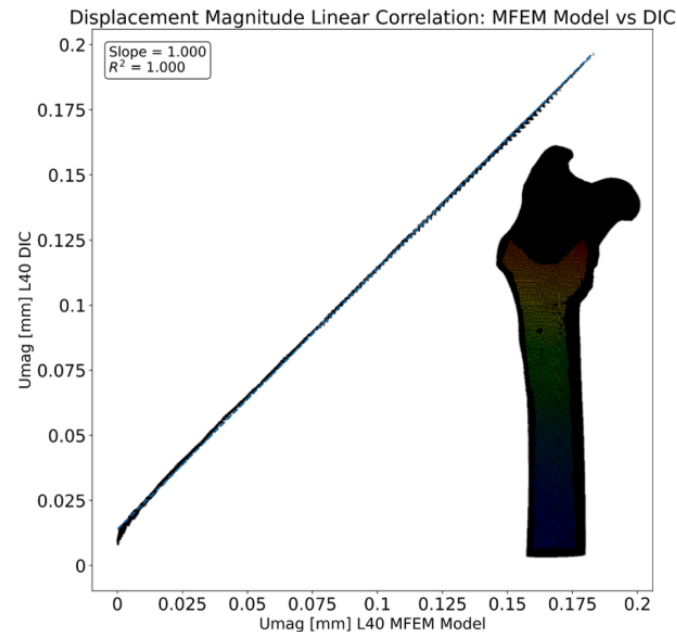
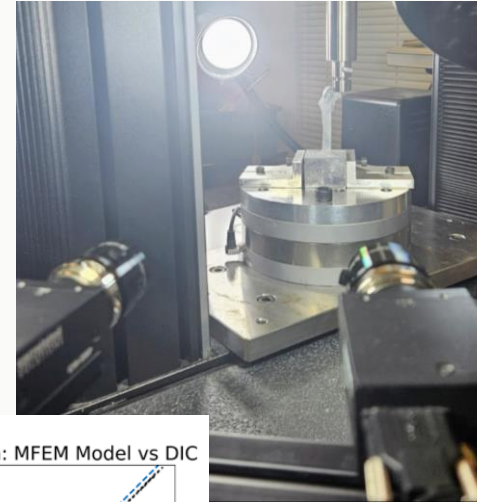


L20 model : displacement magnitude, max. and min. principal strains -

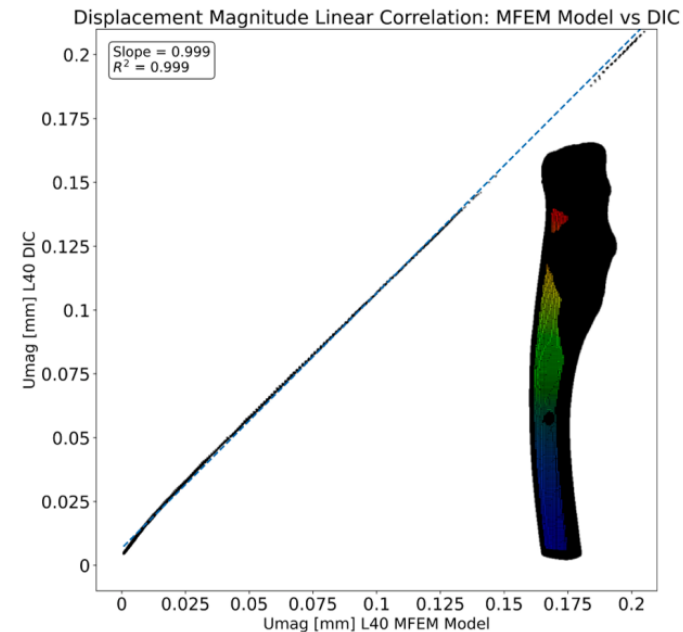
¹ Graph partitioning (Karypis & Kumar) distributes the mesh across MPI ranks for parallel assembly and solve.

Validation with Digital Image Correlation (DIC)

- L40 femur compression test with DIC measurements
- Two independent DIC systems track displacement during loading.
- Comparison of displacement magnitudes between DIC and MFEM shows a strong linear correlation ($R^2 \approx 1.0$)



(a) ROI1 linear correlation.



(b) ROI2 linear correlation.



KEY TAKEAWAYS

What This Means

1. **Extreme scale** : 811M DOFs solved on 192 CPU cores with <700 GB using the matrix-free EBE approach.
2. **Matrix-free multigrid - VMG**: Voxel multigrid greatly reduces assembly/setup cost and memory compared with AMG.
3. **Verified experimentally**: MFEM agrees with Abaqus and DIC.

MFEM Community Workshop

THANK YOU

Questions?

Shani Martinez-Weissberg · sm2@mail.tau.ac.il

Will Pazner · pazner@pdx.edu

Zohar Yosibash · yosibash@tauex.tau.ac.il